

Klorvatan Film-Coated Tablets

Module 3 Quality Documentation — Formulation F-492/A

Facilities and Equipment

Manufacturing is performed at the approved commercial site using dedicated product-contact equipment trains. Equipment qualification and cleaning validation are maintained under the site validation master plan. No changes to the approved facility registration are proposed.

Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results. Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control. Training records for personnel involved in manufacturing and testing are maintained under the site quality system. Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. The control strategy links critical quality attributes to critical process parameters identified during development.

Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control. Cleaning verification swab results were below the health-based exposure limits derived for the product. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable. Hold-time studies established acceptable storage durations for intermediate blends and cores under defined conditions. The control strategy links critical quality attributes to critical process parameters identified during development. Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation.

Process Validation Summary

Process performance qualification was executed on three consecutive commercial-scale batches. All predefined acceptance criteria for critical process parameters and critical quality attributes were met, supporting a conclusion of a validated state of control.

Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles. The control strategy links critical quality attributes to critical process parameters identified during development. Cleaning verification swab results were below the health-based exposure limits derived for the product. No changes to the approved analytical procedures are proposed in this submission.

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Stability Summary

Primary stability studies were conducted under ICH long-term (25°C/60% RH) and accelerated (40°C/75% RH) conditions on three registration batches. All quality attributes remained within acceptance criteria through the tested intervals. Photostability testing per ICH Q1B confirmed that the product is not photolabile in the commercial pack.

Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. Training records for personnel involved in manufacturing and testing are maintained under the site quality system. Cleaning verification swab results were below the health-based exposure limits derived for the product. Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation. The control strategy links critical quality attributes to critical process parameters identified during development.

Analytical Method Validation Summary

The assay and related-substances procedures employ reversed-phase high performance liquid chromatography with ultraviolet detection. Validation demonstrated acceptable specificity, linearity, accuracy, precision, and robustness. Forced degradation studies confirmed the stability-indicating capability of the related-substances method.

The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control.

Hold-time studies established acceptable storage durations for intermediate blends and cores under defined conditions. Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation. Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles. Cleaning verification swab results were below the health-based exposure limits derived for the product.

Reference Standards

Primary reference standards for the drug substance and specified impurities are qualified against compendial standards where available. Working standards are qualified against the primary standard under an approved protocol and are stored under controlled conditions.

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Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable. Training records for personnel involved in manufacturing and testing are maintained under the site quality system. Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation. Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation.

Control of Drug Product

The drug product specification includes tests for description, identification, assay, uniformity of dosage units, dissolution, related substances, water content, and microbial limits. Analytical procedures are compendial where available; non-compendial procedures were validated in accordance with ICH Q2(R1).

Training records for personnel involved in manufacturing and testing are maintained under the site quality system. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable. Statistical evaluation of batch release data demonstrates process capability well within specification limits. No changes to the approved analytical procedures are proposed in this submission.

3.2.P.4 Excipient Control Strategy

For the commercial formulation F-492/A, Pearlitol 200SD (Mannitol, diluent) is used.

Formulation Development Rationale

The formulation development rationale for Pearlitol 200SD rests on improved flowability during high-speed compression.

Grade selection and control follow the principles of ICH Q6A, Ph. Eur. general monograph 2034 and FDA Guidance for Industry: Immediate Release Solid Oral Dosage Forms.

Commercial Control Strategy — Excipients

The current approved commercial specification (SPEC-EX-8883 Rev 05) lists Mannitol as Parreck M200.

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Container Closure System

The commercial packaging configuration consists of high-density polyethylene bottles with child-resistant polypropylene closures and induction seal liners. Container closure integrity was demonstrated by dye ingress testing. The packaging components comply with applicable food-contact regulations and compendial requirements for plastic packaging systems.

The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. Cleaning verification swab results were below the health-based exposure limits derived for the product. Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control. No changes to the approved analytical procedures are proposed in this submission.

Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control. The control strategy links critical quality attributes to critical process parameters identified during development. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable. Cleaning verification swab results were below the health-based exposure limits derived for the product.

Analytical Method Validation Summary

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Statistical evaluation of batch release data demonstrates process capability well within specification limits. Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data

to confirm the continued state of control. Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Cleaning verification swab results were below the health-based exposure limits derived for the product. No changes to the approved analytical procedures are proposed in this submission.

Statistical evaluation of batch release data demonstrates process capability well within specification limits. Cleaning verification swab results were below the health-based exposure limits derived for the product. Deviations observed during the campaign were investigated and closed with no impact to product quality. Training records for personnel involved in manufacturing and testing are maintained under the site quality system.